

R analysis on breast tumor data set; a statistical comparison
MAT2007 - Maastricht university

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Date: October 15, 2026

Introduction

This study investigates whether four selected characteristics for breast tumors (area, symmetry, texture and concavity) differ between the two groups established in the dataset, benign and malignant. Breast cancer is one of the most common among women in the world, therefore its research is of high importance in order to acquire a better understanding of their traits and how they behave in order to, in this case, develop technologies to help predict diagnosis. An initial assessment whether these traits could be a marker to differentiate between the groups is intended. The objective this project has is to answer the following research question: **are the selected characteristics of breast tumors significantly associated with benign or malignant tumor classification?** To provide an answer different statistical parameters were calculated using the programming language R. Among these; the mean, standard deviation (SD), mean difference, standard error (SE) and finally a t test was conducted to determine statistical significance in each trait.

Data sample

The dataset contains around 17.639 numerical data points (each point represents the mean for that specific trait and tumor), however in this project we only used around 2.276 data points for the four traits mentioned in the introduction section. The relevant columns used from the dataset include: ID (tumor identification), diagnosis (M or B), traits (area, symmetry, texture and concavity)

The output is a table displayed in the terminal which contains 8 overall mean values (2 for each trait and tumor group).

Results and discussion

In Figure 1 we can observe that the area and concavity show the biggest difference in their means being 978.3(M)- 462.8(B) for area and 0.16(M)- 0.04(B) for concavity. As well as exhibiting a small overlap in their standard deviation (462.8+/-134.3 (B), 978.4+/-367.9 (M) for area, 0.0460 +/- 0.043 (B), 0.16+/- 0.075 (M) for concavity). Therefore these traits moving in totally different ranges across the two groups makes them stronger as a marker to distinguish malignancy. Meanwhile, symmetry and texture show a greater overlap in the spread of their uncertainties around the mean (error bars) as well as more similar mean which makes the tumors harder to differentiate by these traits

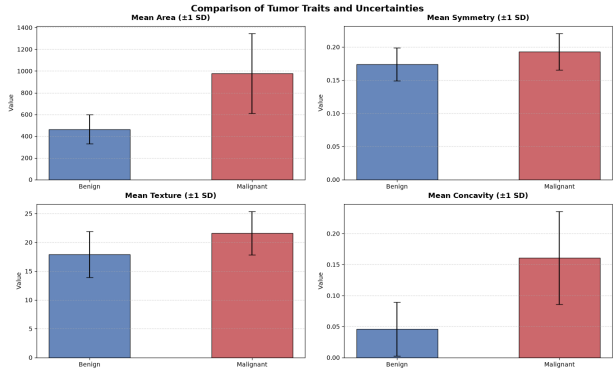


Figure 1: Statistical comparison of the mean vs SD across traits

In Figure 2 we can see that the t-test analyses revealed statistical significant differences between malignant and benign tumors across all four evaluated traits. Because none of the standard error bars reaches the zero reference line all four traits demonstrate significant differences betweent the tumor groups (p is smaller than 0.001)

We can observe that area for malignant tumors is 515 units above bening tumors (SE = 26.5). Texture exibited a mean difference of 3.70 (SE = 0.33), concavity is 0.115 larger in malignant tumors (SE = 0.006). Finally symmetry showed the smallest mean difference across the traits with 0.019 units (SE = 0.002)

The traits are measured in different scales therefore their units have a different magnitude. To see which one fluctuates more, SE would need to be looked at as a percentage of the mean difference.

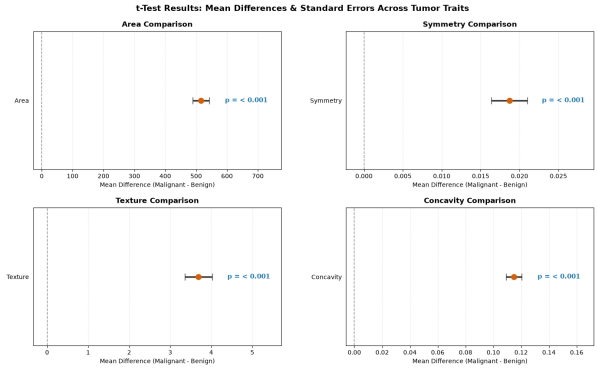


Figure 2: t-Test results; mean differences and their standard errors across the four traits

Conclusion

We can conclude that area and concavity show the strongest differences relative to their uncertainty. However, the results are not a malignancy diagnostic tool, they prove that their significance is not occuring by random chance but actual numerical data supports the statements. Area and concavity could be a marker to distiguish the two groups.

References

[1] W. Nick Street, W. H. Wolberg, O. L. Mangasarian "Nuclear feature extraction for breast tumor diagnosis", Proc. SPIE 1905, Biomedical Image Processing and Biomedical Visualization, null (29 Jul 1993); https://doi.org/10.1117/12.148698